

BRAF fusion-hotspot benchmark report

Study: thca_tcga_pan_can_atlas_2018

Abstract

This report analyzes BRAF gene fusions from the thca_tcga_pan_can_atlas_2018 study, covering 15 fusion events. 15/15 fusions (100.0%) were in-frame. 9/15 fusions (60.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=1$, not statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 100.0% in-frame and 60.0% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 15 analyzed fusion events, identifying 8 distinct fusion partner genes. The most frequent partner was SND1, observed in 5 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 9/15 (60.0%) of in-frame fusions retained the domain, $p=1$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.015984.

Domain disruption

Disruption of the RAS-binding domain and Cysteine-rich domain was tested with Fisher's exact test comparing in-frame fusions against all others; 10/15 (66.7%) of in-frame fusions disrupted the domain, $p=1$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.811419.

Cutpoint detection

Cutpoint detection scanned 15 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 380 aa (exon 8; permutation-corrected $p=0.020979$, statistically significant at $\alpha=0.05$). This is -77 aa from the nearest configured domain boundary at 457 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=9$) and "not_retained" ($n=6$). For Frame_status, retained: 9/9 (100.0%, 95% CI 70.1%-100.0%); not_retained: 6/6 (100.0%, 95% CI 61.0%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, $p=unavailable$.

Composite evidence score

Composite evidence score produced results for this run (8 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association produced results for this run; see the accompanying data for details.

Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

Window detection

Window detection produced results for this run (19 row(s) in window_scan, 6 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SND1	5	0.3333333333333333	0.17963140948233938	0.0090754903114612	0.9832335329341317	0.6552603318391452	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.35973401355691137	1
AP3B1	2	0.13333333333333333	0.17963140948233938	0.0090754903114612	0.9985029940119761	0.6552603318391452	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30202443271858803	2
MACF1	2	0.13333333333333333	0.17963140948233938	0.0090754903114612	0.9985029940119761	0.6552603318391452	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30202443271858803	3
MKRN1	1	0.06666666666666667	0.17963140948233938	0.0090754903114612	0.9970059880239521	0.6552603318391452	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.28179988182038446	4
BCL2L11	1	0.06666666666666667	0.17963140948233938	0.0090754903114612	0.9610778443113772	0.6552603318391452	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2764106602634982	5
FAM114A2	2	0.13333333333333333	0.17963140948233938	0.0090754903114612	0.8248502994011976	0.6552603318391452	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.27597652852697124	6
ZC3HAV1	1	0.06666666666666667	0.17963140948233938	0.0090754903114612	0.8233532934131736	0.6552603318391452	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25575197762876767	7
SUGCT	1	0.06666666666666667	0.17963140948233938	0.0090754903114612	0.0	0.6552603318391452	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1322489836167916	8

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
46	0	9	1	5	0.0	0.4	0.3979400086720376
380	0	9	4	2	0.0	0.010989010989010992	1.9590413923210934
381	5	4	4	2	0.625	1.0	-0.0
393	7	2	5	1	0.7	1.0	-0.0
438	7	2	6	0	0.0	0.48571428571428577	0.3136191229720017

domain_disruption tables

frame_domain_contingency_table

10	0
5	0

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	15	5	0	10	0.0
cysteine_rich	Cysteine-rich domain	15	5	0	10	0.0

domain_retention tables

frame_domain_contingency_table

9	0
6	0

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	15	9	0	6	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	9	15	0.6

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AP3B1	2
BCL2L11	1
FAM114A2	2
MACF1	2
MKRN1	1
SND1	5
SUGCT	1
ZC3HAV1	1

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
181	381	200	8	7	5	4	1.25	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (8 total)
193	393	200	11	4	7	2	1.75	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (11 total)
281	381	100	8	7	5	4	1.25	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (8 total)
293	393	100	11	4	7	2	1.75	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (11 total)
331	381	50	8	7	5	4	1.25	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
343	393	50	11	4	7	2	1.75	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (11 total)
356	381	25	8	7	5	4	1.25	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (8 total)
368	393	25	11	4	7	2	1.75	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (11 total)
380	405	25	11	4	7	2	1.75	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (11 total)
380	430	50	11	4	7	2	1.75	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (11 total)
381	406	25	8	7	7	2	17.5	0.040559440559440565	1.3919080439021245	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, ... (8 total)
381	431	50	8	7	7	2	17.5	0.040559440559440565	1.3919080439021245	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, ... (8 total)
381	481	100	11	4	9	0	None	0.01098901098901099	1.9590413923210936	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, ... (11 total)
381	581	200	11	4	9	0	None	0.01098901098901099	1.9590413923210936	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, ... (11 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
388	438	50	4	11	2	7	0.5714285714285714	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4
389	439	50	6	9	4	5	1.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-ET-A3BN-01-9, ... (6 total)
393	443	50	6	9	4	5	1.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-ET-A3BN-01-9, ... (6 total)
393	493	100	6	9	4	5	1.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-ET-A3BN-01-9, ... (6 total)
393	593	200	6	9	4	5	1.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-ET-A3BN-01-9, ... (6 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
381	481	100	11	4	9	0	None	0.01098901098901099	1.9590413923210936	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, ... (11 total)
381	406	25	8	7	7	2	17.5	0.040559440559440565	1.3919080439021245	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, ... (8 total)
356	381	25	8	7	5	4	1.25	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
368	393	25	11	4	7	2	1.75	1.0	-0.0	EVT-TCGA-DE-A0Y2-01-10, EVT-TCGA-DE-A0Y2-01-5, EVT-TCGA-DJ-A3UZ-01-12, EVT-TCGA-EL-A3T0-01-1, EVT-TCGA-EL-A3T0-01-3, ... (11 total)
388	438	50	4	11	2	7	0.5714285714285714	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4
389	439	50	6	9	4	5	1.6	1.0	-0.0	EVT-TCGA-EL-A3ZK-01-13, EVT-TCGA-EL-A3ZK-01-6, EVT-TCGA-ET-A2MX-01-2, EVT-TCGA-ET-A3BN-01-4, EVT-TCGA-ET-A3BN-01-9, ... (6 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_sitete2_effect_on_frame
EVT-TCGA-EL-A3T0-01-1	TCGA-EL-A3T0-01		AP3B1-BRAF	AP3B1	in-frame	threepri-me	140787584	9	381	False	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA AP3B1-BRAF Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-ET-A2MX-01-2	TCGA-ET-A2MX-01		BCL2L1-1-BRAF	BC L2 L1 1	in-frame	threepri-me	140783157	10	393	False	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA BCL2L1-1-BRAF Fusion	in-frame
EVT-TCGA-EL-A3T0-01-3	TCGA-EL-A3T0-01		BRAF-A3P3B1	AP3B1	in-frame	fivepri-me	140794308	8	380	False	380	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-A3P3B1 Fusion	in-frame
EVT-TCGA-ET-A3BN-01-4	TCGA-ET-A3BN-01		BRAF-FAM114A2	FAM114A2	in-frame	fivepri-me	140783021	10	438	False	438	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-FAM114A2 Fusion	in-frame
EVT-TCGA-DE-A0Y2-01-5	TCGA-DE-A0Y2-01		BRAF-MACF1	MACF1	in-frame	fivepri-me	140794308	8	380	False	380	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-MACF1 Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-EL-A3ZK-01-6	TCGA-EL-A3ZK-01		BRAF-SND1	SN-D1	in-frame	five_prime	140787548	9	393	False	392	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-SND1 Fusion	in-frame
EVT-TCGA-KS-A41I-01-7	TCGA-KS-A41I-01		BRAF-SND1	SN-D1	in-frame	five_prime	140794308	8	380	False	380	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA BRAF-SND1 Fusion	in-frame
EVT-TCGA-FY-A40N-01-8	TCGA-FY-A40N-01		BRAF-SUGCT	SUG-CT	in-frame	five_prime	140924566	1	46	False	46	True	exact match	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		NA BRAF-SUGCT Fusion	in-frame

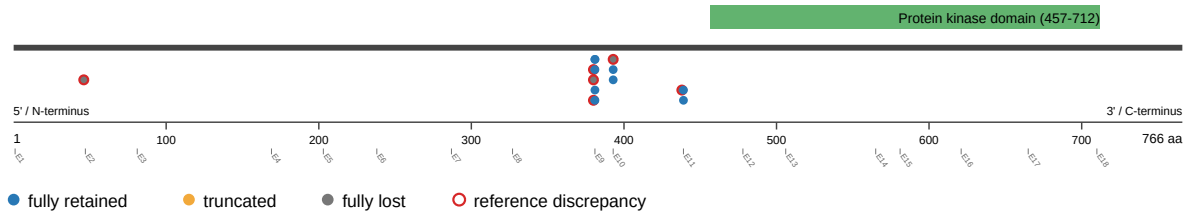
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-ET-A3BN-01-9	TCGA-ET-A3BN-01		FAM114A2-BRAF	FAM114A2	in-frame	threonine_prime	140781693	11	439	False	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA FAM114A2-BRAF Fusion	in-frame
EVT-TCGA-DE-A0Y2-01-10	TCGA-DE-A0Y2-01		MACF1-BRAF	MACF1	in-frame	threonine_prime	140787584	9	381	False	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA MACF1-BRAF Fusion	in-frame
EVT-TCGA-J8-A3O1-01-11	TCGA-J8-A3O1-01		MKRN1-BRAF	MKRN1	in-frame	threonine_prime	140787584	9	381	False	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA MKRN1-BRAF Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-DJ-A3UZ-01-12	TCGA-DJ-A3UZ-01		SN-D1-BRAF	SN-D1	in-frame	there_prime	140787584	9	381	False	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA SND1-BRAF Fusion	in-frame
EVT-TCGA-EL-A3ZK-01-13	TCGA-EL-A3ZK-01		SN-D1-BRAF	SN-D1	in-frame	there_prime	140783157	10	393	False	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA SND1-BRAF Fusion	in-frame
EVT-TCGA-FY-A40N-01-14	TCGA-FY-A40N-01		SN-D1-BRAF	SN-D1	in-frame	there_prime	140787584	9	381	False	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA SND1-BRAF Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-KS-A4ID-01-15	TCGA-KS-A4ID-01		ZC3HAV1-BRAF	ZC3HAV1	in-frame	therapeutic	140781693	11	439	False	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA ZC3HAV1-BRAF Fusion	in-frame

domain retention outliers

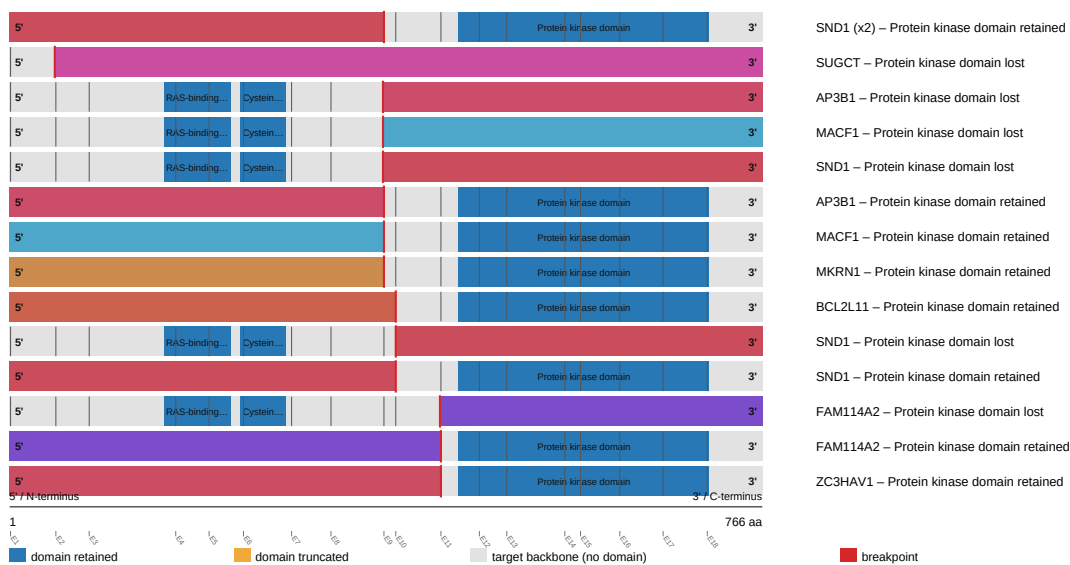
BRAF domain-retention track



fusion schematic

BRAF fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (BRAF portion is domain-colored, exon ticks)



reference comparison

Reference vs thca_tcga_pan_can_atlas_2018

